

8. A bag contains red and blue marbles. When two marbles are drawn, the probability that they are both red is equal to the probability they are both blue. The probability that one of each color is drawn is $\frac{4}{7}$. Find the least number of marbles that are in the bag.
9. The distance between the lines $y = \frac{3}{4}x + b$ and $3x - 4y = 7$ is 2. If $b > 0$, find the value of b .
10. If $i = \sqrt{-1}$, find the sum of an infinite geometric series whose first term is $2 - i$ and common ratio is $\frac{1}{2}i$.
11. Determine the numerical coefficient of $x^2y^2z^2$ in the simplified expansion of $(1+x)^5(1+y)^4(1+z)^3$.
12. The solutions for x in the equation $2(\ln x)^2 = \ln x^5 + 3$ are of the form e^a where e is the base of a natural logarithm. Find the sum of all possible values of a . Write your answer as a common or improper fraction reduced to lowest terms.
13. Given $f(4x+3) = 8x-1$, $f^{-1}(x) = ax+b$. Find the sum $(a+b)$.